insight for Logistics
White Paper

Agentic Control Tower: Conversion to Autonomous Detection-Judgement-Execution Feedback Loop Beyond Visibility

Preview

Over the past decade, the Supply Chain Control Tower (SCCT) has served as a core solution for enhancing visibility across global supply chains. However, despite significant progress in expanding the scope and depth of data available through dashboards, its ability to translate insights into actual execution has remained limited. Powered by Large Language Model (LLM), Agentic AI presents an opportunity to transform the traditional SCCT into an Agentic Control Tower—an autonomous feedback loop of detection, judgement, and execution. By rapidly interpreting a wide range of signals and recommending appropriate response strategies, Agentic AI provides opportunities to advance into Agentic Control Tower.
This white paper examines the evolution of the SCCT, explores the operating principles of Agentic AIalong with leading global use cases, and identifies the key prerequisites for successfully implementing an Agentic Control Tower.

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Professor Sanghwa Song conducts research on digital transformation in logistics and supply chain management (SCM), with a focus on optimization and AI. He has carried out research projects with Incheon Port Authority, Korea District Heating Corporation, Hyundai AutoEver, KOTRA, Korea Expressway Corporation, SK, and Hyundai Steel. He has also served as an advisory role for companies including Naver, CJ Logistics, KOTRA, and Hyundai Steel, as well as a member of the Regulatory Reform Committee of the Korea Ministry of Land, Infrastructure and Transport.

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Contents

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01.

Evolution of Supply Chain Control Tower

Control Tower to Enhance Visibility

The Supply Chain Control Tower (SCCT) is a concept that applies the role of an air traffic control tower, coordinating the aircraft movement, to supply chain management. Just as an air traffic control tower provides a comprehensive view of all aircraft movements on runways and within airspace while coordinating takeoffs and landings, the SCCT integrates and manages fragmented data across the entire supply chain—from procurement and production to logistics and sales—within a single system. It also enables organizations to detect and manage exceptions at an early stage. By consolidating fragmented data into a unified platform, the SCCT facilitates systematic analysis of integrated data, enabling faster decision-making. As global supply chains became increasingly complex and exposed to greater risks since the 2010s, organizations began adopting the SCCT system to enhance end-to-end visibility. In particular, the adoption accelerated significantly during the COVID-19 pandemic, as global companies recognized the importance of securing real-time data visibility.

Unlike the manufacturing industry, where a single enterprise often controls the entire process, the logistics industry is characterized by collaboration among multiple organizations across the end-to-end process. Consequently, logistics-related data is managed across numerous independent systems, making it difficult to manage all information on a single platform. Leading SCCT system providers continuously enhance their systems by proactively identifying, integrating, and processing fragmented data across global supply chains.

For example, project44, a digital platform that provides global supply chain data, has built one of the world's largest logistics databases, connecting more than 259,000 carriers across 186 countries and processing approximately 1.5 billion shipments annually and over 700 million logistics events each day.[1] By consolidating fragmented transportation data into a unified digital platform, gaining real-time visibility into container locations at sea and estimated truck arrival times at port terminals has been enabled. Instead of reaching out to logistics service providers for status updates or waiting until disruptions occurred, companies could monitor transportation activities through a unified platform. This demonstrated that centralized integrated data management alone could significantly improve supply chain visibility.

However, while integrated digital platforms have been effective in digitizing global supply chain flows and collecting broader and richer datasets to a certain extent, they still have limitations in transforming data into actionable insights and operational response strategies. Supply chain professionals must still access the system, manually analyze the necessary information, and determine appropriate response measures. As a result, the industry is increasingly focusing on how to maximize the value created after supply chain visibility has been achieved.

1 AI Magazine, “How Does project44’s AI Agent Redefine Freight Procurement?”, March 2026.1

Agentic AI, Providing New Opportunities

SCCTs have evolved into services that enhance supply chain visibility by presenting relevant data to supply chain professionals. However, in the subsequent stages of issuing early warnings, determining response strategies, and executing operational actions, human judgment and execution capabilities remain essential. Although digital platforms provide real-time access to massive volumes of data, the substantial volumes of data can itself become a burden, making it more difficult for practitioners to make timely and appropriate decisions. To distinguish meaningful signals from noise and identify events requiring immediate attention, supply chain professionals must continuously monitor and analyze the data. The resulting cognitive overload can sometimes reduce operational response capabilities. Conversely, narrowing the scope of monitored data to only a few key indicators may hinder the detection of subtle changes across the supply chain. Consequently, even after overcoming the challenges associated with data collection, the workload imposed on human operators during the decision-making and execution processes remains a critical bottleneck in effective supply chain management.

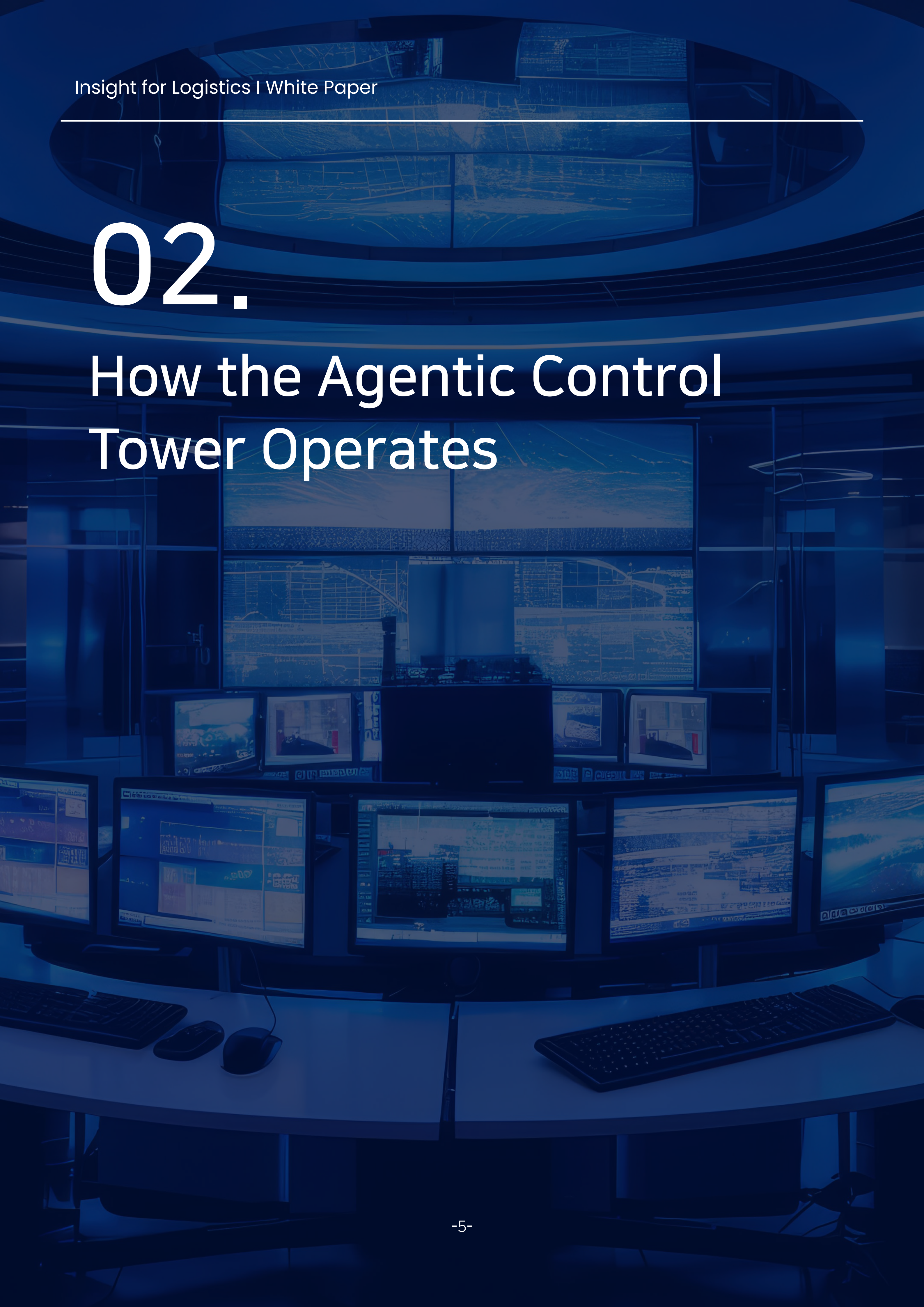
LLM-based Agentic AI is expected to play a transformative role in addressing precisely this bottleneck. Agentic AI refers to autonomous services powered by LLMs, such as ChatGPT, Claude, and Gemini, that operate based on user-defined goals. Unlike conventional software systems operating according to predefined algorithms designed by developers to handle anticipated scenarios, Agentic AI leverages LLMs to understand assigned objectives, autonomously identify the tasks required to achieve them, formulate execution plans, and carry out those plans.[2] Because conventional software operates according to predefined rules and workflows, developers must explicitly specify how the system should respond to each anticipated supply chain scenario in order to address complex situations arising across the supply chain. This rigid design has limited the effectiveness of supply chain control towers in managing the complexity and uncertainty of real-world supply chains. In contrast, Agentic AI offers greater autonomy and flexibility by identifying problems across diverse data sources and generating context-specific response strategies.

Accordingly, an Agentic Control Tower refers to a supply chain control tower that leverages Agentic AI to autonomously perform analysis and execution prior to human intervention, thereby enhancing its operational capabilities. [Table 1] summarizes the key differences between traditional SCCTs and Agentic Control Towers.

2 EY, “Revolutionizing global supply chains with agentic AI”, 2026.

[Table 1] Comparison of Traditional Control Tower and Agentic Control Tower

Category	Traditional Control Tower	Agentic Control Tower (This White Paper)
Key Function	Data integration, Supply chain visibility, Event notification	Detection–Judgement–Execution Feedback Loop
Agent and Function	Human (Dashboard-based data analysis and response)	Share roles between AI agents (autonomous within defined scope) and humans (coordination)
Operation Method	Predefined rule-based logic	Goal-driven autonomous monitoring and decision-making by AI agents
Algorithm	Predefined rules and criteria, Human experience and know-hows	Autonomous utilization of optimization and data analytics tools by LLM-based AI agents
KPI	Data collection scope and frequency, Supply chain visibility level, etc.	Automatic event detection ratio, Detection time, Decision-making cycle, etc.



02.

How the Agentic Control Tower Operates

Autonomous Detection-Judgement-Execution Feedback Loop

In this white paper, an Agentic Control Tower is defined as follows:

“A feedback loop–based supply chain operating infrastructure that collects data across the supply chain in real time, enables AI agents to autonomously detect abnormal situations, formulates response strategies within predefined operational scope, and determines and executes final response actions through collaboration with human operators.”

A key aspect of this definition is the autonomous situational assessment performed by AI agents and their ability to respond within predefined operational boundaries. In traditional SCCT, the entire process—from data collection to alert generation—was executed according to logic predefined by human operators. While this approach was effective for handling known scenarios, it had inherent limitations in responding to today's increasingly complex and unpredictable supply chain risks. By contrast, Agentic AI leverages LLMs as its reasoning engine, enabling it to interpret situations, analyze contextual information, and respond effectively even to previously unseen patterns.

To mitigate potential risks associated with LLM hallucinations and other AI errors, operational guardrails can be predefined to specify the scope and limits within which Agentic AI is allowed to operate. These guardrails ensure that human operators can intervene whenever necessary to assess the situation and determine appropriate response measures.

[Table 2] Evolution of SCCT: From Visibility to Autonomous Execution

Control Tower 1.0 (Supply Chain Visibility)	Control Tower 1.5 (Early Warning)	Control Tower 2.0 (Limited Autonomous Decision-Making; Agentic)
Provides visibility into what’s happening across the supply chain	Identifies supply chain issues and requests that operators develop appropriate response measures	Autonomously performs the Detection -Judgement-Execution feedback loop regarding supply chain issues within predefined operational scope
Dashboard feature and monitoring	Provide early warning and alert, and predefined list of response measures	Develops and recommends response measures within predefined operational scope
Humans are responsible for both issue identification and decision-making	Humans remain responsible for the entire decision-making process	Human intervention is required only for exceptional situations beyond the predefined scope

The concept of the SCCT has gradually evolved from providing supply chain visibility to supporting early warning capabilities. With advances in AI technologies enabling Agentic AI to perform autonomous decision-making, SCCTs are now entering the Control Tower 2.0 era, characterized by limited autonomous responses. Agentic AI continuously analyzes large volumes of data in real time without human intervention, significantly reducing the cognitive burden on supply chain professionals and allowing them to focus on exceptional situations that require human judgment.

Over the longer term, continued advances in AI are expected to extend the role of Agentic AI beyond decision-making into the execution phase. Eventually, many of today's operational boundaries on autonomous decision-making and execution are expected to disappear. As Agentic Control Towers mature to this stage, they will be capable of accessing to enterprise IT systems—including Transportation Management System (TMS), Warehouse Management System (WMS), and Enterprise Resource Planning (ERP) system —to update operational data and execute approved response actions automatically. Once this level of technological maturity is achieved, the detection–judgement–execution feedback loop will become fully autonomous, realizing a truly autonomous SCCT.

Shifting KPI: From Decision-making Quality to Speed and Scope

Until now, the expected primary objective of supply chain control towers has been to improve the quality of problem-solving and decision-making. Because human operators were responsible for the entire detection–judgement–execution process rather than simply monitoring data, enhancing the quality of decisions supported by the control tower became the central measure of its effectiveness.

In contrast, the emergence of the Agentic Control Tower is expected to shift from the quality of individual decisions to the speed and coverage of decision-making. Under conventional control tower operations, human operators were required to monitor complex situations and develop response strategies within limited time and resource constraints. Consequently, they had little choice but to focus on the most critical issues. As a result, improving decision quality became the primary objective of control tower enhancement. Agentic AI, however, is not constrained by cognitive overload and is capable of making autonomous judgments even in exceptional situations that fall outside predefined analytical scenarios. As a result, both the scope and the speed of decision-making can be significantly expanded. In the past, improving decision-making speed and extending its scope required assigning additional personnel. Because the availability of human resources was constrained by economic considerations, both the operational scope and the response speed of supply chain control towers inevitably remained limited. Agentic AI fundamentally changes this paradigm by enabling the collection and rapid analysis of massive volumes of data along with the formulation of response strategies even when previously unseen exceptions emerge. Nevertheless, the quality of individual decisions may still be constrained by the capabilities of the underlying LLM. Consequently, human intervention may remain necessary in particular situations.

Capability to Process Unstructured Data

Agentic AI's capability to process unstructured data—text, images, video—plays a major role in expanding the scope and speed of decision-making within the Agentic Control Tower. Traditional supply chain control towers operated by collecting and analyzing structured data organized in databases according to predefined criteria, or by relying on numerical data gathered from automated equipment and sensors. In the case of unstructured data, it was essential for a human to manually interpret its meaning and convert it into structured data.

However, problem situations that arise in the field are often difficult to analyze and respond to using only predefined structured data. It is not easy to capture every possible problem situation in a predetermined format in advance, and even when a situation is similar to a known case but has slight exceptions, structured data analysis often struggles to identify it. Since Agentic AI operates based on LLMs, the capabilities of the underlying LLM are critical. Current LLM-based AI services can analyze not only text data but also images and video, and even within text data, they can grasp not only information organized in table form but also details that are hard to detect within complex, unstructured text.

The range of data types needed to identify exceptional situations is extremely diverse: emails from carriers about container shipment delays, messages from customs authorities, news articles about port strikes, photos of damaged container trucks, and even phone calls reporting such incidents. In the past, for significant problems with ripple effects, separate technologies were sometimes developed specifically to convert unstructured data into structured data. What differentiates the Agentic Control Tower is that Agentic AI possesses the built-in capability to interpret and judge unstructured data without requiring any such separate technology.

Supply Chain Knowledge: Ontology

For the Agentic Control Tower to process unstructured data and operate its detection judgement execution feedback loop, analysis and judgment of supply chain data are essential in the LLM based decision making process. In this context, data handling methods such as ontologies play a pivotal role. LLMs like ChatGPT, Claude, and Gemini are trained on massive amounts of internet text, images, and video, giving them broad AI capabilities. However, their ability to address internal knowledge that exists only within a specific industry can be limited. Industry specific terminology, knowledge required for particular technologies, and internal expressions unique to a company reside in closed systems and cannot be used for LLM training. Consequently, a generic LLM may struggle with such data.

When this happens, generic LLMs can produce hallucinations—making incorrect judgments or offering seemingly plausible but inaccurate recommendations. If errors accumulate, users will lose trust in the Agentic AI's decisions and suggested actions. Even though a universally designed LLM—often called a "foundation model"—can handle most tasks efficiently, there remains a need for technologies that supply the model with industry and enterprise specific knowledge. Ontology becomes the key to leveraging Agentic AI in this regard.

An ontology can be viewed as a structured map of the relationships among particular entities and the rules or constraints governing those relationships. For example, if the context "An order is transported by one or more containers; container transport is executed under a contract; the contract links cost and penalty" is encoded as a map in an ontology, an LLM can refer to it while operating within the detection judgement execution feedback loop. When a newly received email mentions "penalty cost," the ontology enables the system to understand that penalty is connected to order, container, and contract.[3]

Because ontologies encapsulate contexts that are specific to each individual supply chain or company, building an ontology adds an extra step when constructing an Agentic Control Tower. Nevertheless, this step is indispensable for assigning site specific services to a generally purposed LLM.

3 Supply Chain Management Review, "Execution, not chat: How Agentic AI changes supply chain operations", February 2026.

03.

Agentic Control Tower Use Cases of Global Companies

Project44: Evolving from a Supply Chain Platform to Agentic AI Operating System

Project44, which has focused on collecting and sharing supply chain data, is actively pushing its transition toward becoming an Agentic Control Tower. In April 2026, project44 announced that its portfolio of AI agents—coordinating core freight operations such as freight procurement, disruption management, network operations, exception management, and slot booking—has enabled shippers to reduce freight costs, improve inventory turnover, and manage supply chain risk more effectively.[4]

Project44 identified the reason existing AI agents have failed in real-world deployments not as a shortcoming of the AI models themselves, but as a lack of strong operational grounding in the field. In contrast, project44 announced that it has built an integrated architecture that leverages data and knowledge accumulated over a decade from the world's largest real-time logistics network, deep domain expertise delivered to each agent, and the deployment of the right agent at the right time. This underscores that the true differentiator in applying Agentic AI lies in field-specific data and context.

Blue Yonder: AI Agents that Aid Users to Act

Blue Yonder, a provider of SCM software, unveiled its "Cognitive Solutions"-based SCM offering in May 2026, adopting extensive integration of AI agents. According to announcements made at Blue Yonder's own conference, ICON 2026, the company launched solutions embedding AI agents across each domain of the supply chain—including inventory, supply chain planning, warehouse operations, and logistics management—providing proactive support in which AI agents analyze data and provide alternatives for users using the SCM software.[5] These AI agents were introduced as enabling predictive event management, automated root-cause analysis, and AI-powered search and answers across each supply chain sector.

According to Blue Yonder, the adoption of AI agents reduced WMS migration timelines by 87%, from 11 weeks to 7 days, and cut supply chain data collection time from two weeks to just 8 hours. For a major North American beverage distributor operating a complex, multi-tier supply chain network with roughly 326,000 demand forecasting points, Blue Yonder reported that the company successfully transitioned its entire forecasting and replenishment engine to an AI agent-based system in just 72 hours.[6] Blue Yonder has long supplied integrated SCM software built around the concept of a supply chain control tower—managing supply chain data on a single dashboard and enabling users to make rapid decisions through domain-specific solutions. The fact that Blue Yonder has published numerous cases where AI agents dramatically shortened the time required to feed into users' decision-making feedback loops illustrates how the core KPI of the Agentic Control Tower is shifting from decision quality to scope and speed.

4 project44, "project44 launches AI agent portfolio at decision44, built on a decade of context, skills, and orchestration", Press Release, April 8, 2026.

5 Blue Yonder, "Blue Yonder Launches New Cognitive Solutions and AI-Driven Innovations at ICON 2026", Business Wire, May 19, 2026.

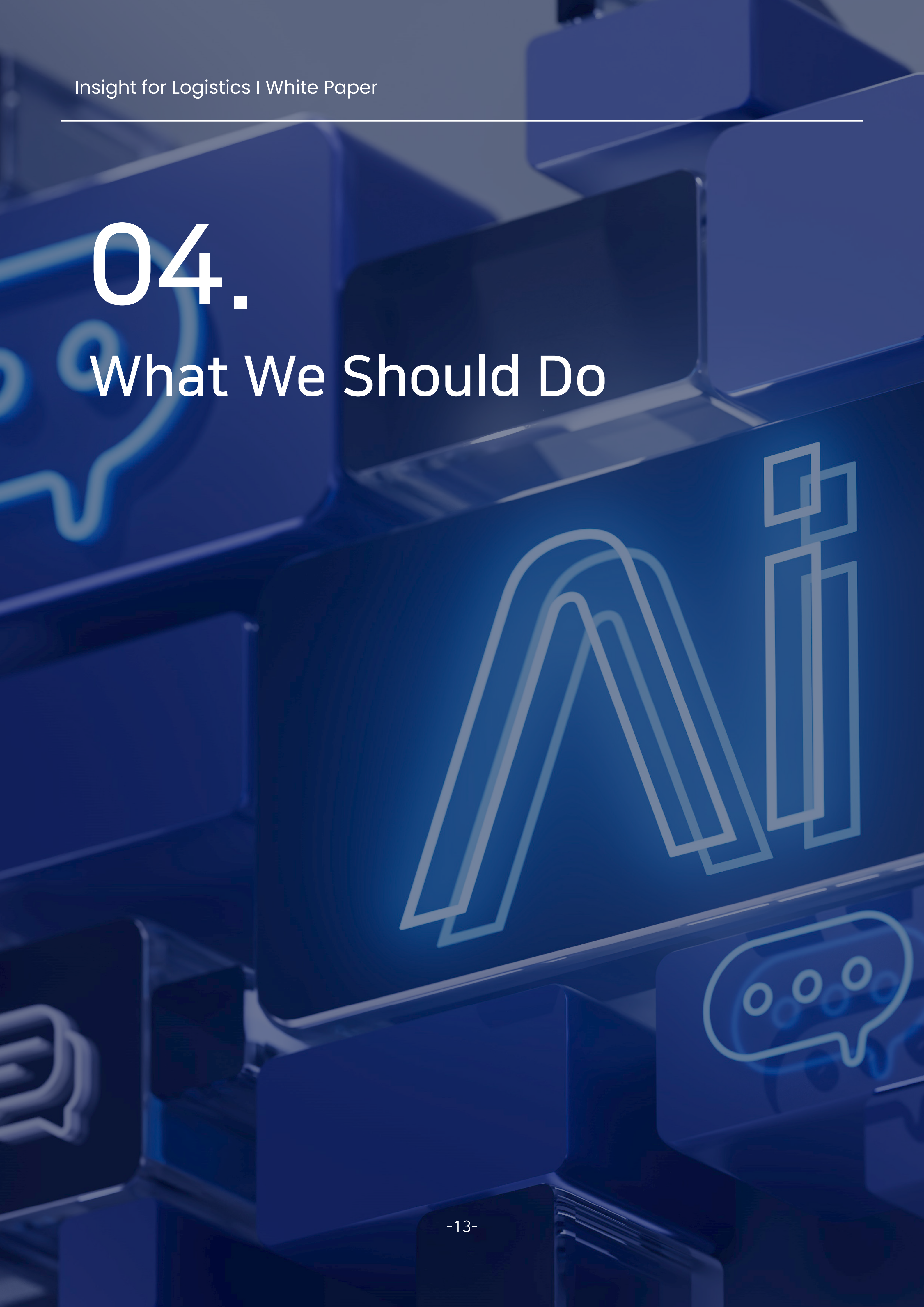
6 diginomica, "ICON 2026 : Blue Yonder bets everything on Cognitive", May 2026; Blue Yonder, "Supply Chain Control Tower Software", blueyonder.com.

Net Feasa: Connecting AI Agents to Containers

Net Feasa, an Irish digital SCM solutions company, unveiled an AI platform that has trademarked the very name "Agentic Control Tower."[7] Net Feasa's AI platform is built around transforming the ocean container itself into an AI agent: an AI agent is attached to every container in transit, and each container-level agent connects to a marketplace that supports ocean freight negotiation and optimal bidding, creating a structure that enables optimal container transport. The intriguing aspect is that negotiation and bidding tasks—traditionally handled by international logistics forwarding experts—are now carried out autonomously by the AI agent linked to each container. When AI agents are deployed in negotiations and drive optimal decisions, the speed of decision-making is expected to improve significantly.

7 Net Feasa, "Net Feasa unveils Agentic Control Tower™ novel AI Platform", EIN Presswire, May 12, 2025.

04.
What We Should Do



What We Should Do

As we have seen, the Agentic Control Tower represents infrastructure that goes beyond the mere collection and monitoring of data—beyond "a better integrated dashboard showing more diverse data" or "fancier visualization"—to advance the decision-making structure for supply chain operations based on AI agents. The criteria for evaluating a supply chain control tower are now shifting from "how much diverse data can it collect?" and "how effectively can it visualize data?" toward "how much has the speed of the detection-judgement-execution feedback loop increased?" and "how broad is the scope over which the feedback loop is applied, rather than visualization?" From this perspective, the tasks companies must prepare for can be summarized as follows.

First, securing data and building an ontology is more urgent than adopting AI agents. A review of global cases and the fundamental characteristics of AI agent technology shows that while AI agents, powered by LLMs, can be deployed across a wide range of tasks, the success of adoption ultimately depends on how effectively field-specific knowledge and know-how are conveyed to the AI agent. As emphasized in project44's announcement, given the inherent limitations of general-purpose LLMs, the success of the Agentic Control Tower hinges on how effectively the AI agent can understand the closed, real-world work environment. The key is to provide large-scale data effectively enough for the AI agent to operate, and to use ontologies to describe the relationships between data points and supply the contextual meaning of the data.

Second, it is important to start from small, limiting the scope of AI agent application to high-frequency tasks where the impact of failure is low. As discussed earlier, AI agents are still likely to struggle with understanding field data. Unlike structured data, if AI agents fail to properly process the vast amount of unstructured data present in the field, their detection-judgement-execution feedback loop is likely to malfunction. Accordingly, it is important to first deploy AI agents in a limited way for high-frequency tasks—where failures carry low impact and occur often enough to enable learning—in order to achieve small wins.

Third, the core KPI for the supply chain control tower should be scope and speed, not quality. Transitioning a supply chain control tower into an Agentic AI-based Agentic Control Tower means increasing scope and speed so that a wide range of scenarios that might previously have been missed can now be detected and addressed proactively. Therefore, in the process of adopting an Agentic Control Tower, setting a goal such as "supporting a higher level of decision-making than before"—one framed around quality—is likely to fail. The essence of Agentic AI lies in catching opportunities that people might miss, responding quickly, alerting the responsible personnel to the event, and proposing a course of action. This resolves the problem of cognitive overload for personnel and creates an environment in which they can focus on the events that truly matter.

Over the past decade, the logistics industry has invested in systems to improve supply chain visibility, driving digital transformation across the sector as a whole. The foundation is now in place for AI agents to be deployed in connecting the data secured through digital transformation to the detection-judgement-execution feedback loop. While AI agents trained on the vast, general data of the internet excel at general-purpose capabilities, they remain weak in logistics- and supply chain-specific training data—which is precisely why the infrastructure and data secured through digital transformation are expected to provide significant support for AI adoption. The Agentic Control Tower is expected to play a major role in the journey from digital transformation to the broader AI transformation.

Samsung SDS monitors and responds to logistics risks worldwide in real time through Global Control Center (GCC).
This illustrates supply chain visibility and operational capabilities that underpin the Agentic Control Tower presented in this white paper.

「Samsung SDS's Global Logistics Risk Management Solution」

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